

- **Symbolic Grounding:** Each image in a Bongard-LOGO problem (positives, negatives, and query) is represented as a symbolic input—either a procedural grammar string or a descriptive natural-language version—using its ground-truth program.
- **Textual Reasoning:** A language model receives only the symbolic inputs and must infer a rule and classify the query as positive or negative. No pixel data or visual features are provided.

This setup transforms the task from multimodal perception-plus-reasoning to a purely symbolic inference problem. Crucially, it allows us to isolate the role of input representation in driving model behavior. If LLMs perform well in the symbolic setting but VLMs fail on raw images, this pattern points to a representational or grounding bottleneck rather than a lack of reasoning capacity.

4 Experiments

We design our experiments to answer three questions: (i) do symbolic representations unlock abstract visual reasoning that VLMs fail to exhibit on Bongard-LOGO, (ii) how does the *form* of symbolic input (programs vs. descriptions, with or without concepts and images) modulate performance, and (iii) do models rely on *rule-like structure* rather than exploiting token-frequency or lexical-overlap heuristics?

Models. We evaluate proprietary Gemini models (Google DeepMind, 2024), using Gemini-2.5-Flash as the primary end-to-end visual baseline and including Gemini-3-Flash in the supplementary model-capacity analysis. We also evaluate open-source models including DeepSeek-R1 (8B, 14B, 32B) (Guo et al., 2025), Phi-4 and Phi-4-Reasoning (14B) (Microsoft Research, 2024), Qwen2.5 and Qwen3 (up to 32B) (Yang et al., 2025), Gemma 3 (12B, 27B), and Mistral Magistral. Open models are run locally using the Ollama framework. Refer to Appendix B for more details.

Evaluation Protocol. Each Bongard-LOGO problem contains 6 positive examples, 6 negative examples, and 1 query image (detailed in Appendix A). Models must predict whether the query belongs to the positive or negative class. Accuracy is computed over a fixed 2,000-problem subset containing 500 problems from each reported evaluation

split: BD, FF, HD-Comb, and HD-Novel. The subset is drawn uniformly at random from the test split without any performance-based filtering, and all models are evaluated on the same subset for comparability. Open-source models use deterministic decoding (temperature 0.0, no sampling); Gemini models use low-temperature official modes. Outputs are parsed from a constrained JSON format containing a reasoning trace, induced rule, and binary label; unparsable outputs are counted as errors. Exact prompts, model identifiers, and infrastructure details appear in Appendix C,D.

4.1 Main Conditions: From Pixels to Symbols

Our primary comparison contrasts an end-to-end visual baseline with variants of the Componential-Grammatical (C-G) interface, where models operate on symbolic input only.

Visual VLM Baseline. Gemini-2.5-Flash receives all 13 *images* and directly predicts the query label. This represents a standard end-to-end VLM pipeline in which perception and reasoning are tightly coupled.

C-G (Formal Grammar). Models receive only procedural shape grammar strings (as shown in Appendix A.3) derived from the ground-truth LOGO programs for all 13 images, with no pixel input. This condition tests whether explicit, compositional structure alone enables high-accuracy rule induction and classification.

C-G (Natural Language). Instead of grammar strings, models see step-by-step English descriptions of the drawing actions and geometric relations as seen in Appendix A.4. This probes whether recasting the same structure in familiar linguistic form helps or harms abstract reasoning relative to a compact formal syntax.

4.2 Ablations: How Much Structure and Supervision?

We next vary the amount and form of guidance around the symbolic input.

Minimal-Context Grammar. Models receive raw grammar strings plus a brief instruction describing the task, but no explanation of the formalism itself. This few-shot setting asks whether models can infer a novel syntactic interface and the intended rule-induction task from examples alone.

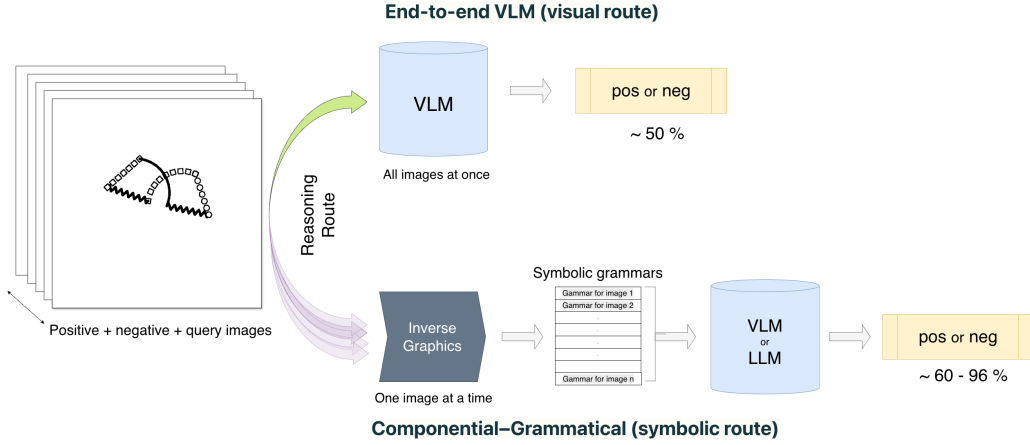


Figure 2: Comparison of visual and symbolic pipelines on Bongard-LOGO. In the *visual route* (top), a Vision-Language Model (VLM) receives images directly and must infer both structure and rules, typically yielding near-chance accuracy. In the *symbolic route* (bottom), problem instances are represented by LOGO-style action programs; a language model then performs rule induction and classification using these symbolic representations, achieving substantially higher performance on the same task.

Concept-Conditioned C-G. The high-level target concept (e.g., “is convex”, “contains a triangle” Appendix A.5) is provided in the system prompt alongside the action programs. This converts the task from rule discovery to rule verification and tests whether explicit concept supervision provides additional gains over purely example-based learning.

Grounded C-G. Models receive symbolic programs for all 13 images plus the *rendered query image*. This condition probes whether a single visual anchor, tied to otherwise symbolic input, yields further improvements beyond the purely symbolic setting. Because it requires image input, we restrict this variant to VLMs that accept both text and images (Gemma3:12B, Gemma3:27B, LLaVA-Llama3, and Qwen2.5-VL:32B).

4.3 Structure vs. Surface: Token Randomization Tests

Finally, we test whether models rely on genuine rule learning over symbolic structure rather than superficial token matching.

Randomizing Query Programs. In the first control, we randomly permute the sequence of drawing actions in the *query* program while keeping the support examples, the query image, and its label unchanged. If models primarily exploit token inventory alone, performance should remain similar; if they rely on structured comparisons between query and support examples, accuracy should degrade once the ordered composition of the query is

broken.

Shuffling Positive and Negative Sets. In the second control, we randomly reassign the 12 support examples between the positive and negative sets while leaving the query program, query image, and query label untouched. This preserves within-problem token inventories but destroys the original support-set partition. Performance therefore need not fall exactly to 50%: after shuffling, the support sets can still instantiate alternative regularities that partially align with the unchanged query. The diagnostic signal is the drop relative to the unperturbed Action Program condition.

We report these controls only for BD and FF. HD performance is already comparatively low in the unperturbed setting, so the same perturbations would be less informative there.

Together, these perturbation studies test whether high C-G performance depends on relational and compositional structure in the symbolic input, rather than on simple token-frequency heuristics alone.

5 Results

We report findings from systematic evaluation of the Componential-Grammatical (C-G) paradigm across 12 models and multiple input regimes. Our core result is consistent: when models receive structured symbolic input in place of raw perceptual input, performance improves dramatically, especially on conceptually grounded reasoning tasks.

Table 1: Summary of key results across experimental conditions on Bongard-LOGO. The first five rows and the randomization rows are mean accuracy (%) across 12 language models. Rows marked with [†] are averaged over the matched 4-model VLM-capable subset (Gemma3 12B/27B, LLaVA-Llama3, Qwen2.5-VL 32B) and should be compared only within that subset. ‘Base’ denotes the minimal-context symbolic baseline. AP = Action Program (formal LOGO-style grammar), AD = Action Description (natural-language description of actions).

Condition	FF	BD	HD
Visual VLM (Gemini-2.5)	50.2	49.8	50.1
C–G (AP)	78.1	68.8	61.0
C–G (AD)	79.3	72.0	59.1
C–G + Concept (AP + C)	79.3	70.0	61.1
Minimal-Context Grammar	77.3	67.7	59.6
Visual baseline (VLM subset) [†]	62.6	57.1	57.5
Grounded C–G (AP) [†]	62.4	55.3	58.2
Grounded C–G (AD) [†]	61.2	55.3	55.8
Category Permutation	70.6	60.1	–
Sequence Permutation	58.0	64.2	–

5.1 Main Effects of Symbolic Grounding

Table 1 reports mean accuracy across three benchmark categories and key experimental conditions. Gemini-2.5-Flash achieves chance-level performance (50–51%) across the board when operating on raw images. In contrast, when provided with symbolic input (Action Programs), models reach up to 96.2% accuracy (Phi-4-Reasoning, Free-form), with average gains of roughly 20–30 points on BD and FF and smaller, but still substantial, gains on HD over the visual baseline.

These results confirm that models’ poor performance in the visual setting stems primarily from representational limitations rather than an inability to reason over structured input. The pooled symbolic conditions improve markedly over the visual-only Gemini baseline, and the matched four-model subset shows that adding a single query image does not recover additional gains once the same models already receive symbolic structure. Across conditions, FF is easiest to recover from symbolic input, BD is intermediate, and HD remains the most challenging.

Beyond overall accuracy, we also examine how models treat positive vs. negative examples under the AP interface. Across datasets, accuracy on positives substantially exceeds accuracy on negatives, and a chi-square test confirms a strong dependence of correctness on example class (see Appendix E.7 for details). This indicates that models adopt a

liberal decision criterion that favors “positive” classifications: they are more reliable at recognizing configurations that satisfy a learned rule than at rejecting heterogeneous counterexamples.

5.2 Descriptive vs. Procedural Input

When symbolic structure is rendered in natural language, pooled accuracy is higher than AP on FF and BD, while AP retains a modest advantage on HD. Figure 3 makes the direction of that difference more explicit: relative to the minimal-context symbolic baseline, AD yields larger mean gains on FF and BD (+2.0 and +4.3 points), whereas on HD the mean AD shift is slightly negative (−0.6) and AP remains modestly positive (+1.3). This pattern is not driven by a single outlier model: the point clouds show that AD is broadly competitive on the easier FF and BD regimes, whereas AP is slightly more stable on the human-designed regime (Table 1).

5.3 Concept Conditioning and Minimal Guidance

Providing models with the target rule label (AP + C) yields modest but heterogeneous effects. Relative to AP alone, the mean change is +1.3 points on FF, +1.2 on BD, and near zero on HD (+0.2), but Figure 3 shows that these averages mask mixed model-level behavior rather than a universal improvement. The appendix tables clarify why: some larger models show positive average gains under concept conditioning, whereas smaller or more instruction-sensitive models exhibit mixed or even negative deltas (Tables A.4 and A.6). We therefore treat AP + C as a model-dependent prompt variant rather than a uniformly stronger condition. The same pattern appears in the minimal-context setting: a shorter prompt can outperform the full prompt for some models, suggesting that additional guidance sometimes introduces distraction or prompt overload rather than useful supervision.

5.4 Visual Grounding Offers Little Added Benefit

Within the matched VLM-capable subset, adding a single visual anchor in the Grounded C–G setting leaves accuracy essentially unchanged relative to the corresponding symbolic-only baseline and slightly below the same subset’s visual baseline on FF and BD (Table 1; see Appendix E.5). The main result is therefore not that query grounding helps a little, but that once symbolic structure is already